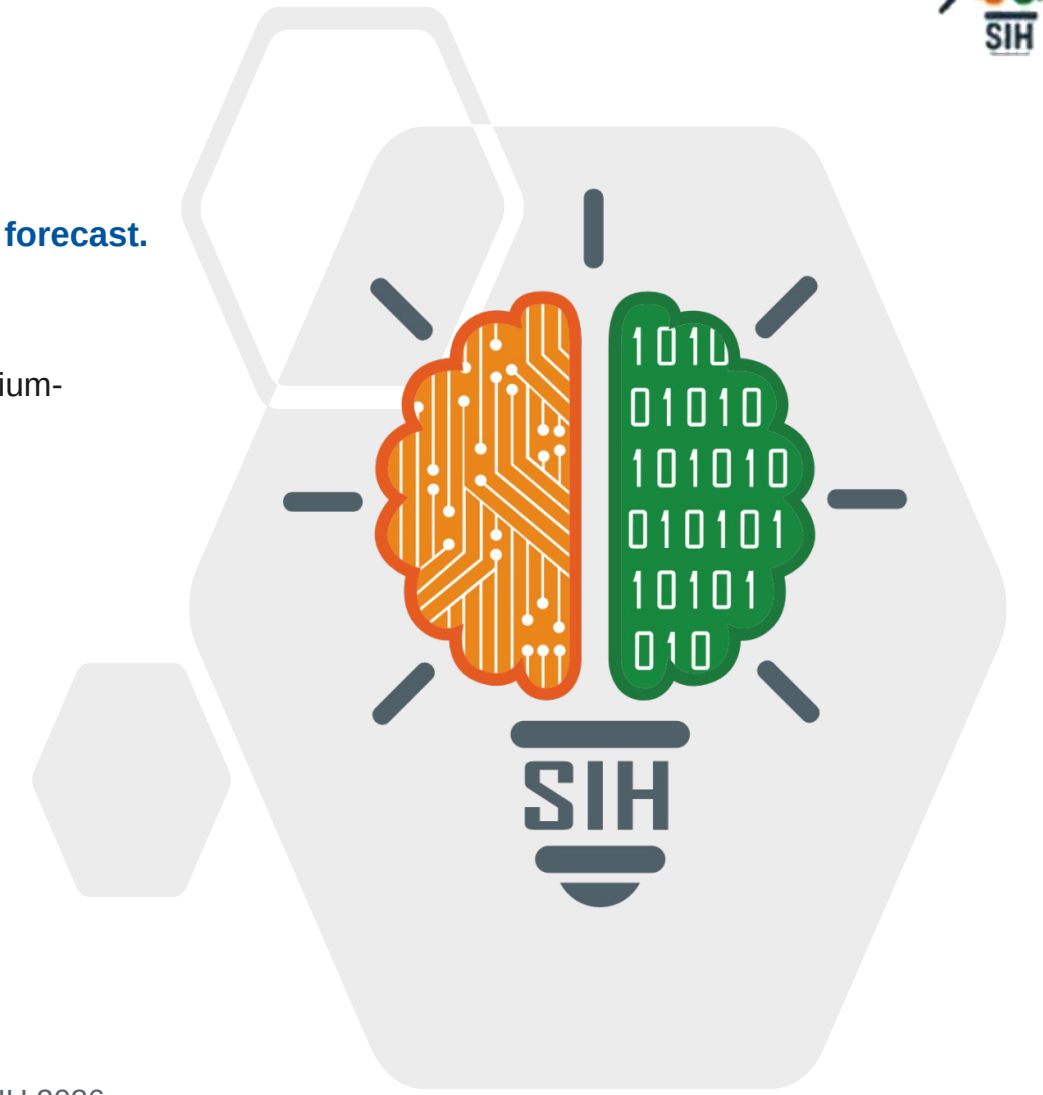
SMART INDIA HACKATHON 2026

VISHWAS

Forecast Confidence Intelligence. Knowing when not to trust the forecast.

- **Problem Statement ID:** SIH26079
- **Problem Statement Title:** AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:** [paste Team ID from SIH portal]
- **Team Name:** Hellfire Labs



VISHWAS · FORECAST BUST DETECTION

VISHWAS is a layer on top of the forecast, not another forecast

It scores every rainfall forecast by location and lead time, Day 1 to Day 10.



It flags busts. It does not change the forecast.

HOW IT ADDRESSES THE PROBLEM

A duty forecaster assesses confidence today by **eyeballing ensemble spread**, by hand, on every forecast that goes out. And the forecast underneath decays fast: **GEFS detects 23% of heavy-rain events at Day 1 and 0.2% by Day 10**.

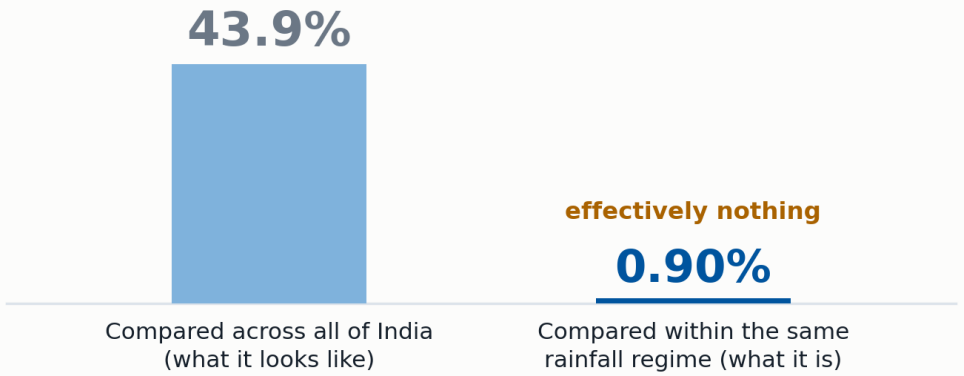
INNOVATION AND UNIQUENESS

Measured against India's own gauge observations, spread appears to explain **43.9%** of the variance in forecast error, but **under 1%** once you compare forecasts in the same rainfall regime.

We are automating a judgement that currently has no evidence behind it.

Ensemble spread explains far less than it appears to

Share of the variance in forecast error that ensemble spread explains.



IMD gauge-observed rainfall. 12 forecasts, four seasons, 2012 / 2015 / 2018.
The within-regime figure is the most generous of five wet-point definitions; the range is 0.17% to 0.90%.

AND WE BUILT THE REPLACEMENT

On that same fair comparison ensemble spread scores **0.565**. VISHWAS scores **0.728**, a margin of **+0.13 to +0.16**, with a 95% interval entirely above zero, on a test year the model had never seen.

TECHNICAL APPROACH

1 FETCH
01_fetch.py
GEFS v12 reforecast, public S3

→

2 FEATURES
02_features.py
11 members → mean + spread

→

3 TRUTH
imd_truth.py
IMD gauge rainfall, exact colocation

→

4 LABEL
06_label.py
bust = error above 95th pct, train only

→

5 MODEL
07_model.py
HistGradientBoosting, then calibration

→

6 EVALUATE
08_met_metrics.py
vs spread, stratified, + SHAP

LANGUAGE & CORE
Python 3 · NumPy · pandas · Parquet

DATA ACCESS
NOAA public S3 · xarray + cfgrib · imdlib

MODEL
scikit-learn HistGradientBoosting

EXPLAIN & VERIFY
SHAP · permutation · cluster bootstrap

OUTPUT
matplotlib · confidence map per lead day

ALL OPEN SOURCE · NO GPU · NO PAID API · NO CREDENTIALS

DATA: ALL PUBLIC, NO COST

GEFSv12 reforecast ensemble supplies every predictor. **IMD 0.25° gauge-gridded rainfall** is the truth: a gauge observation, not a reanalysis, so no model physics is shared with the forecast being judged.

What the model sees

23 features, every one from the ensemble forecast. None from the truth.

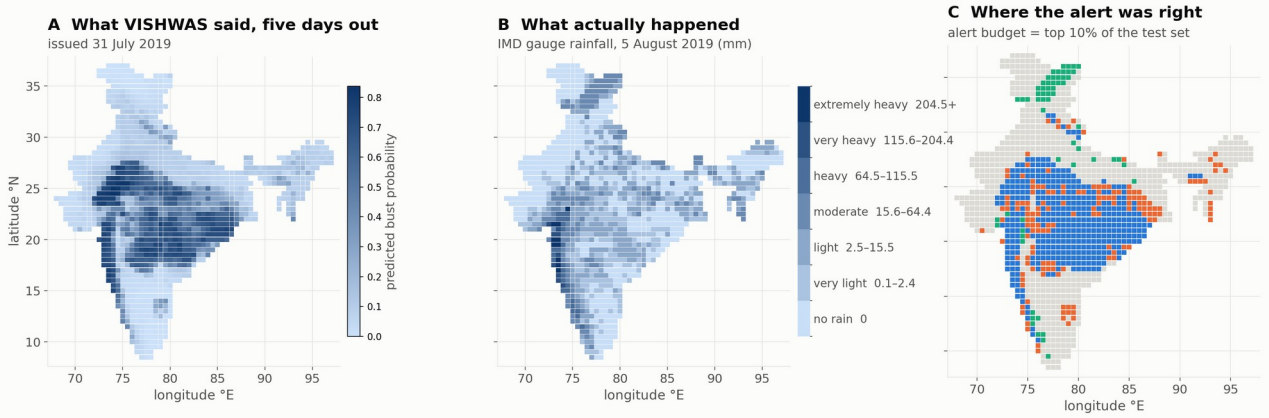


Six of the twenty-three measure where the 11 members disagree with each other.
That disagreement is the whole of the method forecasters use today. Here it is one input among twenty-three.

A bust is |forecast − observed| above the 95th percentile, per region and lead time. Remove the rainfall forecast and the score rises, 0.728 to 0.731.

VISHWAS flagged central India five days before the rain came

One forecast date, Day 5. 454 of 1,252 gauge-observed land gridpoints busted — the highest bust rate of the 52 test dates.



Truth: IMD 0.25 deg gauge-gridded daily rainfall, colocated by exact float match onto the 0.5 deg GEFS grid. Bust = |ensemble mean − observed| above the 95th percentile for that region and lead time, fitted on 2018 training data only. Alert threshold p ≥ 0.153, fixed once on the whole 2019 test set (651,040 rows, 52 forecast dates) and applied here unchanged — not re-fitted for this date. On this date: precision 0.72, recall 0.87, n = 1,252 gridpoints.
ONE DATE, CHOSEN AS AN ILLUSTRATION OF THE PRODUCT'S FORM — not evidence of general performance. The general result is a stratified ROC-AUC of 0.728 against 0.565 for ensemble spread, over all 52 test dates.

FEASIBILITY AND VIABILITY

The forecast already exists. The data already exists. We closed the gap, and measured it.

FEASIBILITY: THE REPLACEMENT WORKS

Trained on 2018, tested on all 52 Wednesdays of 2019, a year the model never saw.

On the fair, within-regime comparison ensemble spread scores **0.565** and VISHWAS scores **0.728**. A margin of **+0.13 to +0.16**, 95% interval entirely above zero, ahead in all eight rainfall bins.

Same alert budget: VISHWAS catches **64% of busts against 55%**.

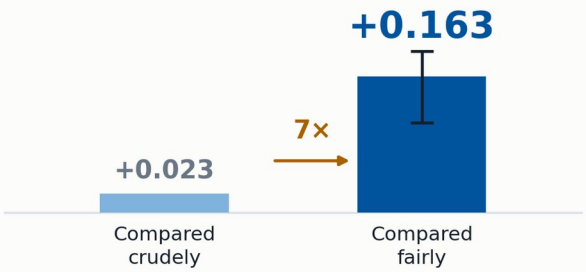
Three defects in our own work were found and fixed: two data leaks, and a tie in the baseline.

VIABILITY: COULD THIS RUN AT NCMRWF TOMORROW?

Features come from ensemble fields any operational centre already produces. CPU only, no GPU. **50.3 GB and 1.3 million gridpoint-days on a laptop.** A layer over an existing run, not a new one.

What the crude comparison hides

Two ways the incumbent method looks better than it is.



VISHWAS's margin over ensemble spread (ROC-AUC). Compared fairly it is seven times larger. Pooling lets ensemble spread track how hard it is raining, and hands it skill it never earned.

189,758 forecasts where ensemble spread is exactly 0.00 mm

29% of the 651,040 forecast test set

One tie the incumbent cannot rank.

VISHWAS ranks inside it.

POTENTIAL CHALLENGES AND RISKS, AND THE STRATEGY FOR EACH

WHY GEFS, NOT NEPS

NEPS archives are not public. GEFS is the only medium-range ensemble with a public 20-year reforecast, and we move to NEPS when access exists.

LAND-ONLY DOMAIN

The IMD gauge grid covers 1,252 of 5,041 gridpoints. That is the gauge domain, and we state it ourselves.

ONE TEST YEAR IS THIN

The interval resamples 52 dates inside one year. Treat it as a floor, not a ceiling. Extending to five years needs no new code.

FALSE ALARMS COST MORE

Busts are ~6% of cases, so we report PR-AUC, never accuracy. An assistant that cries wolf gets switched off, so we tune toward precision.

IMPACT AND BENEFITS

Automating a judgement that currently has no evidence behind it.

WHO THE SYSTEM SERVES

Primary: the duty forecaster at an operational centre such as NCMRWF or IMD, where confidence is assessed by hand on every cycle.

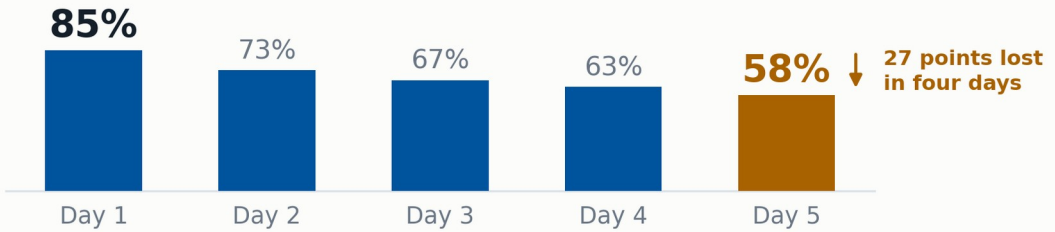
Downstream: district authorities, agricultural advisories, power scheduling and aviation, all acting on the same forecast.

The chart alongside is the problem owner's own number. Skill falls with lead time, and that is exactly the window VISHWAS scores.

IMD's own heavy-rainfall verification reports a false-alarm ratio alongside these accuracy figures.

IMD's warning accuracy falls with lead time

Correct heavy-rainfall warnings, Day 1 to Day 5.
This is exactly the window VISHWAS scores.



Ministry of Earth Sciences, Lok Sabha Q. 2953, 17 December 2025.
The Ministry reports these five figures directly.

PRIMARY FIELD RESEARCH: WE ASKED THREE PEOPLE WHO ACT ON THE FORECAST

FARMER · ASSAM

Shifted a sowing date for rain that never arrived. Lost seed, labour and time.

KRISHI VIBHAG · UNNAO, UP

Called advance knowledge of forecast reliability "very, very beneficial" for planning ahead.

NDRF CONSTABLE · GUWAHATI

Has mobilised for forecast heavy rain that never came.

BENEFITS OF THE SOLUTION

OPERATIONAL

Confidence becomes computed and reviewable, not judged by eye.

ECONOMIC

Fewer people and vehicles committed to forecasts that were never reliable.

SOCIAL

Everyone acts on the same forecast, with an honest reliability label attached.

"Don't just know the forecast. Know whether to trust it."

RESEARCH AND REFERENCES

OUR OWN MEASUREMENT

12 GEFSv12 reforecasts spanning four seasons and three years (2012 / 2015 / 2018), 77,425 wet samples over 1,252 gauge-observed land gridpoints, Day 1 to 10, scored against IMD gauge-gridded rainfall.

Ensemble spread appears to explain **43.9%** of the variance in forecast error. Controlling for rainfall magnitude it explains **0.17% to 0.90%**. The range spans five different definitions of a “wet” gridpoint, so the finding does not rest on one arbitrary cut.

The learned layer, trained on 2018 and tested on 2019, scores **0.728** on that same fair comparison against ensemble spread’s **0.565**, measured on the 651,040-row 2019 test set drawn from a 1,302,080-row labelled table.

Uncertainty is assessed with a cluster bootstrap over forecast dates, because the independent unit is the forecast, not the gridpoint. The range above spans analyst choices, not sampling error.

DATA SOURCES

- **NOAA GEFSv12 Reforecast.** Public bucket s3://noaa-gefs-retrospective/ registry.opendata.aws/noaa-gefs-reforecast
- **IMD 0.25° Gridded Daily Rainfall, 1901 to 2024.** imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html
- **Guan et al. (2022).** GEFSv12 reforecast dataset. Mon. Wea. Rev. 150(3). doi.org/10.1175/MWR-D-21-0245.1

WHO DID THIS WORK: AN INTERDISCIPLINARY TEAM

B.Tech: Devansh (data pipeline, model and evaluation) · Ankit (problem-statement tracking, option analysis)

BCA: Arnav (forecast-failure case studies) · Paras (metrics and evaluation integrity)

BBA / BA: Mahika (stakeholder interviews and pitch) · Dhruv (users, impact and sourcing)

BACKGROUND AND VERIFICATION PRACTICE

- **India Meteorological Department.** Monsoon Report 2024. Heavy-rainfall verification: false-alarm ratio, probability of detection and threat score. mausam.imd.gov.in
- **India Meteorological Department.** Annual Report 2024, district-level forecast coverage. mausam.imd.gov.in
- **Ministry of Earth Sciences.** Assessment of the economic benefits of weather and climate services. moes.gov.in
- **Central Water Commission.** Flood Management in India: A Statistical Report, 2023. Background on the decisions that ride on a medium-range forecast. cwc.gov.in
- **Ashrit et al. (2020), NCMRWF.** Prediction of the August 2018 heavy rainfall events over Kerala with high-resolution NWP models. Meteorological Applications 27(2). doi.org/10.1002/met.1906
- **Kumar et al. (2008).** Analysis of the 26 July 2005 heavy rain event over Mumbai. Q. J. R. Meteorol. Soc. 134. Source of the Santacruz / Colaba contrast. doi.org/10.1002/qj.325
- **Ministry of Earth Sciences.** Lok Sabha Q. 2953, 17 Dec 2025. Heavy-rainfall warning accuracy of 85, 73, 67, 63 and 58% at Day 1 to Day 5, and ~40% improvement on 2014. moes.gov.in/sites/default/files/2953English.pdf
- **LeClerc & Joslyn (2015).** The Cry Wolf Effect and Weather-Related Decision Making. Risk Analysis 35(3). Why an over-warning assistant loses its users. doi.org/10.1111/risa.12336
- **Kusunose & Mahmood (2016).** Imperfect forecasts and decision making in agriculture. Agricultural Systems 146. doi.org/10.1016/j.agsy.2016.04.006
- **WMO / Jolliffe & Stephenson.** Forecast Verification: standard definitions for POD, FAR, CSI, ETS and FSS. doi.org/10.1002/9781119960003